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Sarcasm in political discourse: comparative analysis of YouTube comments on Trump and Biden

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Introduction

Nowadays sarcasm is becoming increasingly popular in online political debates, particularly on social media, where public discourse is often informal, emotional, and highly polarized (Mejova & Srinivasan, 2012; Bitecofer, 2020; Woods et al., 2024). While first serving as a way to make jokes or express humour of a person, it is now used as a powerful tool to criticize political figures, ridicule opposing viewpoints, and define the group affiliation, which commonly leads to radical splits towards political poles and increased negativity among counterpartisans (Nguyen et al., 2021; Mc Cambridge, 2022). As a result, sarcasm can influence the perception and interpretation of information and sometimes even change the viewpoint of the audience (Oprea & Magdy, 2020).

The political significance of online communication was particularly evident following the Cambridge Analytica scandal, which revealed how social media data could be used to manipulate political views and voter behavior through targeted messaging (Heawood, 2018; Arora & Zinolabedini, 2019). While much of the discussion is focused on data-driven political advertising, less attention has been paid to the role of everyday online language. Sarcasm, in particular, can subtly portray political figures as incompetent, dishonest, or absurd, and therefore influence audience attitudes without direct impact (Suwami et al., 2025; Alcaraz-Mármol & Almela, 2024).

Given these observations, it is of crucial importance to investigate how sarcasm operates within concrete political contexts. This study examines sarcasm in online political discourse using YouTube comments on videos of Donald Trump and Joe Biden, making it therefore well-suited for comparative analysis of opposing political parties. Additionally, this study is particularly relevant for scholars in political communication, computational social science, and media studies who want to understand how linguistic strategies shape political attitudes in online environments. Quantitatively, we investigate the audience responses to analyze how supporters and opponents use sarcasm to respond to political content.

Using Zero-shot classifier and roBERTa models, the analysis reveals the frequency and contextual patterns of sarcastic remarks, as well as compares which ML method detects sarcasm more accurately. In addition, K-means clustering method is applied to classify the videos by topics to showcase which global matters attract the audience to express sarcasm the most. This approach makes it possible to investigate whether certain political topics are more strongly associated with sarcastic discourse and whether these associations vary among the presidential

candidates. Against this background, the present study addresses the following research question: **“Which political topics are most strongly associated with sarcastic discourse, and how do they differ per presidential candidate?”**

Literature review

Understanding sarcasm as a comprehensive mechanism influencing individual and collective opinions requires prior knowledge of the parties' division in political settings and its rapid share adjustment.

Political polarization has become a central feature of contemporary democratic politics, particularly in the United States (Bitecofer, 2020; Woods et al, 2024). Bitecofer (2020) confirms that the 2020 U.S. presidential election between Joe Biden and Donald Trump was characterized by extreme levels of negative partisanship. He states that political behavior was primarily motivated by hostility toward the opposing party, rather than by evaluations of candidates' policy, positions or performance. Woods et al. (2024) build on this perspective by demonstrating that polarization is closely linked to identity formation, particularly through nationalist rhetoric, which both candidates used to win the votes. While Trump primarily relied on an exclusionary and grievance-based form of nationalism, Biden emphasized a more inclusive and civic understanding of national belonging. As a result, political discourse increasingly took on a symbolic and emotional function, in which language is used to affirm group membership and delegitimize political opponents.

Within this context, rhetoric plays a crucial role, as it enables speakers to communicate effective opposition while applying indirect communicative strategies, such as mockery, ridicule, and especially sarcasm. Oprea and Magdy (2020), demonstrate that sociocultural factors shape how sarcasm is used and understood online, suggesting that sarcastic messages can subtly affect audience attitudes.

In online political discourse, sarcasm functions not only as a humorous device but also as a political strategy. Mejova and Srinivasan (2012) show that YouTube comments represent a significant form of political discourse, in which users actively respond to political content and interact with one another. In contrast to traditional media, YouTube allows for immediate, visible, and often confrontational forms of engagement. Recent studies further demonstrate that political content on YouTube is not neutrally organized, but ideologically structured. Lai et al. (2024) reveal that political YouTube videos can be systematically positioned along ideological

dimensions. Particularly, current research suggests that sarcastic discourse is likely to cluster around specific, emotionally or symbolically salient political topics rather than being evenly distributed across all forms of political communication. Likewise, McCambridge (2022) demonstrates that sarcasm and mockery in YouTube comments frequently operate as socially acceptable forms of verbal aggression. By framing criticism as humor or irony, users can attack political opponents without appearing overtly hostile. Therefore, YouTube provides a well-suited framework for investigating how sarcasm varies across political topics and between candidate-centered discussions.

Due to the implicit and context-dependent nature of sarcasm, it is challenging to automatically detect this form of language through simple methods. In their computational-based linguistic approach, Kunneman et al., 2015 use supervised machine-learning classifier trained on Dutch Twitter data to detect sarcasm beyond explicit lexical markers. They reveal that sarcasm is rarely identifiable through individual words alone but instead relies heavily on context and shared knowledge between speaker and audience. However, it is often characterized by polarity reversal and exaggeration, which causes traditional sentiment analysis approaches to perform poorly.

Nevertheless, recent advances in natural language processing have led to more effective computational approaches to sarcasm detection. Razali et al. (2021) demonstrate that deep learning models incorporating contextual features significantly outperform earlier approaches. Transformer-based models are particularly effective in this regard, as they model meaning in relation to the full contextual structure of a sentence. RoBERTa is an example of such a transformer-based language model and is known for its strong performance on tasks involving subtle linguistic phenomena (Liu et al., 2019). In the domain of sarcasm detection, transformer-based approaches have achieved high accuracy, including in shared evaluation tasks such as SemEval-2022 (Du et al., 2022). These findings support the use of RoBERTa for analyzing sarcastic language in informal political texts.

A key methodological challenge, however, is the lack of large-scale, manually labeled sarcasm datasets. Annotating sarcasm is time-consuming and highly interpretive, particularly in political contexts. To address this limitation, Yin et al., (2019) employed a zero-shot classification approach. They ascertained that zero-shot text classification can be achieved by reformulating labels as natural language hypotheses and using models trained on natural language inference. Rather than replacing supervised classification, zero-shot approaches can complement

transformer-based models by enabling exploratory analysis in contexts where labeled data is scarce. Zero-shot approaches therefore make it possible to analyze sarcastic discourse at scale without extensive manual annotation, making this method particularly suitable for social media data.

Although existing research highlights the importance of sarcasm in online political discourse and emphasizes the role of it in affective polarization, several questions remain unanswered. First, there is limited research that explicitly examines how sarcasm varies across political topics within a single platform. Second, relatively little attention has been paid to systematic comparisons between candidate-centered models and its best suitability depending on language platform. Particularly, it remains unclear how the performance of the models differ depending on linguistic norms, local styles and common rules of a platform (e.g. comment length, meme culture or degree of informality), which can differently influence the expression and recognition of sarcasm phenomena. Finally, advanced computational approaches to sarcasm detection have rarely been applied to large-scale YouTube data.

To address these gaps, the present study suggests a novel comparative approach in identifying sarcasm using advanced Transformers models and combines K-means topic analysis with its correlation to sarcastic YouTube comments. Having said that, we derive three essential objectives for our study:

1. Evaluate statistical performance of RoBERTa and Zero-Shot models of sarcasm detection in videos.
2. Compare sarcasm score in Joe Biden and Donald Trump context.
3. Identify which political topics attract the most sarcastic and mocking comments in online discussions using K-means.

Data

Data preparation

This study analyzes two datasets of YouTube comments associated with videos featuring Joe Biden and Donald Trump. Since the data was extracted through publicly available YouTube platform, no extra sensitive or personal information was incorporated, posing minimal ethical risk to individuals.

After merging the data banks and constructing the additional variables, such as *President* label and *Video Title* collected from YouTube API, resulting dataset consisted of 36,231 comments, from which 20,500 comments (56.58%) were associated with Trump-related videos and 15,731 comments (43.42%) from Biden. Both candidates' comments are well-balanced, allowing for meaningful comparative analysis.

Data preprocessing

As part of the data preparation process, all comments were preprocessed to reduce noise while preserving linguistic features relevant for sarcasm detection, and all titles were tokenized for topic clustering. Comments with missing text were removed, and automatic language identification was applied to keep only English texts, resulting in the exclusion of approximately 20% of the data. Basic noise such as URLs, HTML tags, timestamps, and standalone numeric strings was filtered out. Since emojis are also considered as an important feature for sarcasm detection, they were converted into textual descriptions rather than removed, and common character encoding artifacts were corrected to improve text readability. A limited set of political abbreviations(e.g., *POTUS*, *MAGA*, *GOP*) was expanded for better semantic interpretability, while informal expressions were intentionally preserved to maintain the naturalistic characteristics of YouTube discourse. Additionally, excessive punctuation was normalized and the low-quality and spam-like comments were excluded to retain expressions in a readable manner. Furthermore, the video titles were extracted separately from the YouTube API and deliberately pre-processed only minimally (tokenization) to preserve their natural wording and enable the identification of thematic patterns in the data. Having said that, the final dataset was ready to use for sarcasm detection and topic clustering.

Methods

To align with the **first objective**, we needed to estimate which model handles sarcasm detection most accurately. That is why first 500 random rows from the original dataset were selected and labelled manually using a binary indicator (1 if the comment is sarcastic, 0 otherwise). *Table 1* demonstrates the distribution of sarcastic comments per 500 observations.

Table 1. Distribution of manually labelled sarcastic comments

Sarcastic label	Count	Percentage (%)
0	373	74.6
1	127	25.4

Next, for the Zero-Shot classifier pre-trained “facebook/bart-large-mnli” model was used. Input texts were taken from the *comment_cleaned* column of the evaluation dataset (stored as “eval_500_clean.csv” for code reproduction). The model was run in a binary zero-example configuration, with “sarcastic” and “not sarcastic” as candidate labels. Comments were processed in mini-batches (batch size = 32) to improve efficiency. For each comment, the model provided a probability value of the comment being sarcastic for each observation, which would further on be used to estimate the top-5 most-least sarcastic comments as a ground-truth check. Additionally, *confidence* column showcased how certain the model was towards the binary outcome assigned. This column is useful to support robustness checks and do a quick visual comparison; however, the main analysis still relies on binary classification, not how confident the model is. The final prediction was hereafter turned into a numeric variable and stored into the column *is_sarcastic*. Overall, all the resulting columns were integrated into the original dataset without removing any existing variables.

With regards to roBERTa algorithm, a different approach followed. In general, the model decomposes each comment into contextual tokens using a *transformer mask* and *tokenizer* that shows the relationships between the words in the sentence. This way instead of relying on individual keywords, the model identifies discrepancies between literal meaning and underlying intent to characterize sarcasm. Classification then occurs by assigning the comment to the category that best fits this contextual representation (Du et al., 2022).

With respect to the model execution, firstly the *text_for_model* column stored a copy of *comments_cleaned* pillar to not cause any invertible changes. In the new column we reverted every comment value to a string format, collapsed multiple spaces into one and removed leading and trailing spaces. After texts were cleaned and filtered, only non-empty posts with valid binary sarcasm labels remained. In the “Sarcastic” column we ensured numeric format. Afterwards, the labeled subset was divided into training (20%), development (40%), and test (40%) sets to obtain class proportions and to avoid using pre-trained algorithm on completely new data. The model was refined as a binary sequence classifier with a learning rate of 1×10^{-5} , a batch size of 16, and three training epochs. Prior to adjusting these configurations, we tested multiple fine-tuning approaches, such as modifying the number of training epochs, reducing the learning rate, experimenting with different train-validation splits, and applying overall calibration, which is thoroughly explained in the **Results** section. After training, the model was applied to all available 400 comments, generating both binary predictions and sarcasm

probabilities. In a similar manner to the Zero-Shot experiment, these predictions were appended to the original dataset and saved for further analysis.

Lastly, models performance was evaluated using confusion matrices, accuracy, precision, recall and F1-score.

For Objective three K-means clustering was used to aggregate videos in groups based on video title and the comments content. Predicated upon the elbow rule, the algorithm suggested ten different clusters (*Fig.1*); however, cluster distribution revealed several groups containing only one video. That is why it was decided to cut down the number of clusters to five. In addition, a *post-hoc semantic interpretation of clusters* was applied as part of a descriptive method to suggest the possible interpretation of video clusters. The underlying idea of this technique is in cluster classification based on the average TF-IDF values of terms occurring in the corresponding video titles. Fundamentally, for each cluster, the top-three terms from the titles with the highest values of TF-IDF were extracted to obtain an initial thematic classification (Saclain et al., 2016).

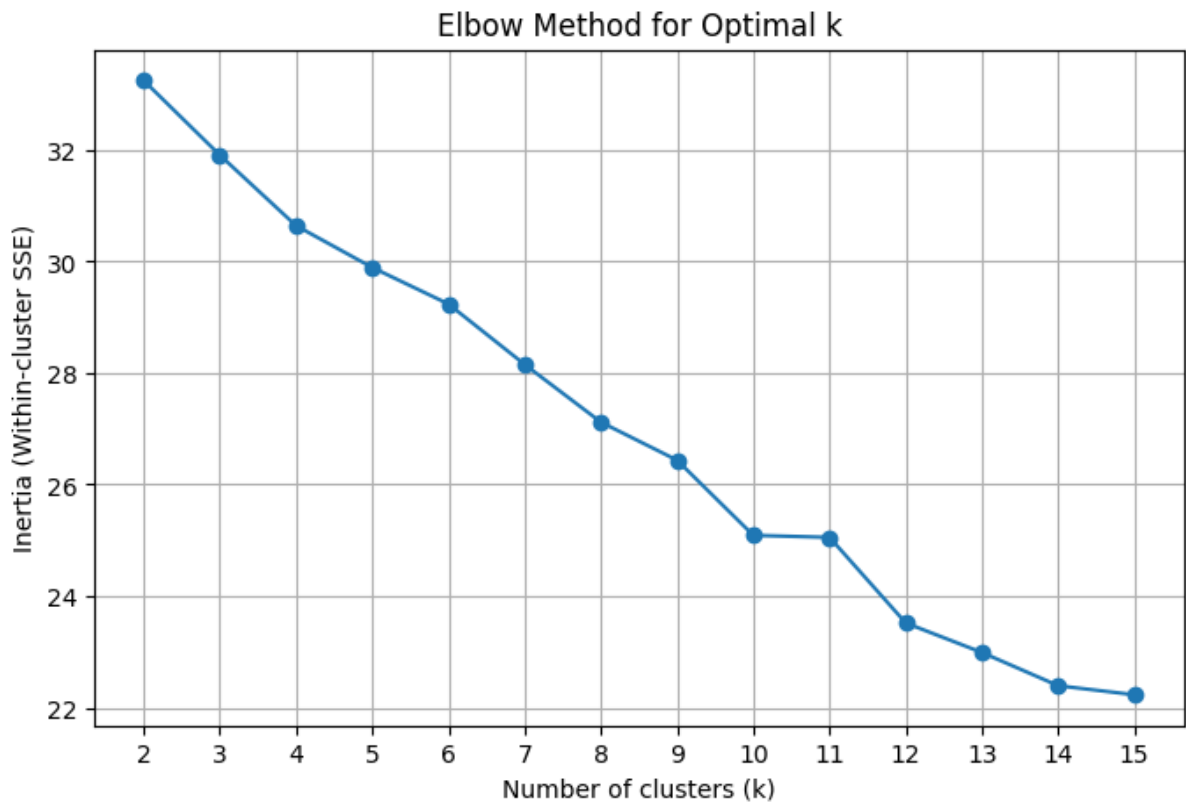


Figure 1. Elbow method identifying suitable number of clusters

Results

This section presents the performance results of each classification algorithm in the experiment using all combined feature sets. Table 3 reflects the score metrics from both models, the results are presented in terms of F1 score, precision, recall, and accuracy.

Table 2. Score metrics of Transformer models

Model	Accuracy	Precision	Recall	F1
RoBERTa	0.746	0.000	0.000	0.000
Zero-shot	0.716	0.448276	0.511811	0.478

Looking from an overall perspective, it is evident that even though the RoBERTa model demonstrates higher overall accuracy (0.746), it failed to detect sarcastic remarks completely, resulting in zero precision, recall, and F1 scores. This indicates that under these conditions, the supervised model fails to distinguish sarcasm cues. As mentioned previously, several fine-tuning methods were applied to improve the model performance, from switching the number of epochs to decreasing the learning rate; however, no successful results were achieved. This is assumed to be true due to the initially limited number of labeled examples, quickly followed by overfitting. Additionally, even the decreasing validation loss does not indicate successful learning (Appendix A), confirming a lack of generalizability to unknown data.

In contrast, the zero-example model achieves slightly lower accuracy (0.710), but significantly higher precision (0.44), recall (0.52), and a higher F1 score (0.48), highlighting the ability of this method to detect sarcastic comments. This illustrates the limitations of accuracy as a single criterion in imbalanced classification.

Figure 2 (a-b) demonstrates the performance of both models in the confusion matrix in comparison to actual sarcastic comments.

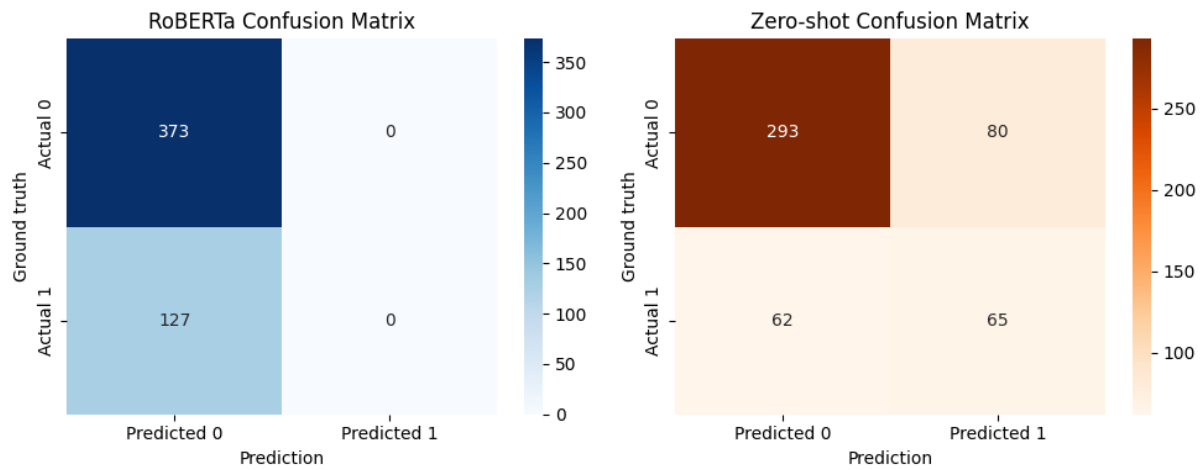


Figure 2 (a-b) RoBERTa and Zero-shot confusion matrices

Overall, roBERTa is only effective if the text contains sufficient contextual information (Du et al., 2022). It also required a lot more time for running predictions on the sample dataset, fine-tuning the training arguments, splitting the test/dev/train in different proportions and calibrating with a “brute force” to achieve the better outcome. Having said all that, it was decided to proceed with a zero-shot classifier for labelling all the data in a dataset.

Following Zero-shot automated labelling, out of 28,268 comments, 8,185 (29.0%) were classified as sarcastic, and 20,083 (71.0%) as non-sarcastic. The average confidence level of the model was 68.82%, indicating reasonable trust in the comment classification. To establish a qualitative foundation, the five comments with the highest and lowest confidence level for sarcasm were extracted (Fig 3-4). As judged by human inspection, we agree with algorithm identification and find the extracted comments reasonable.

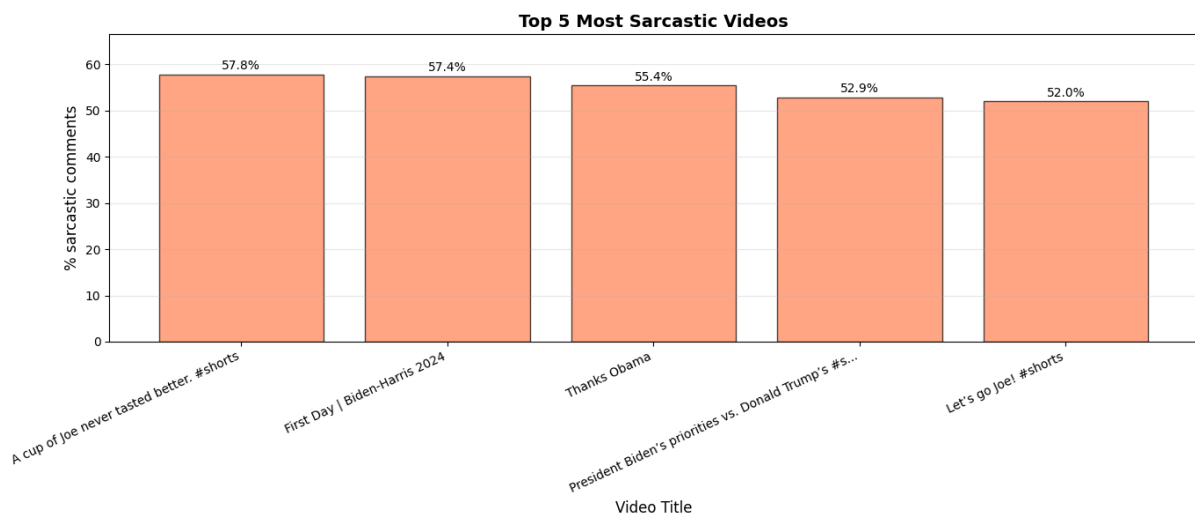


Figure 3. Top-5 ranked videos with highest sarcastic score

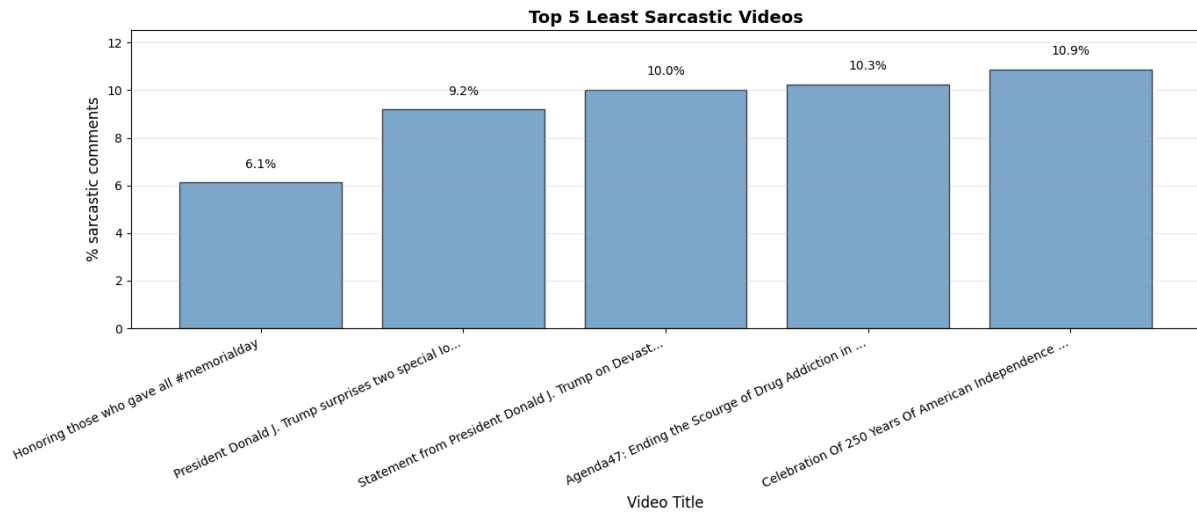


Figure 4. Top-5 ranked videos with lowest sarcastic score

Figure 5 reveals the share of sarcastic comments across Biden and Trump videos. Answering the **second objective**, videos of Biden constitute a higher share of sarcasm than comments under videos of Trump. This pattern may partially reflect the large number of memes that emerged in response to the president's achievements, as well as the resulting weakening of his public talks (Ross, 2021; Tassadit, 2025).

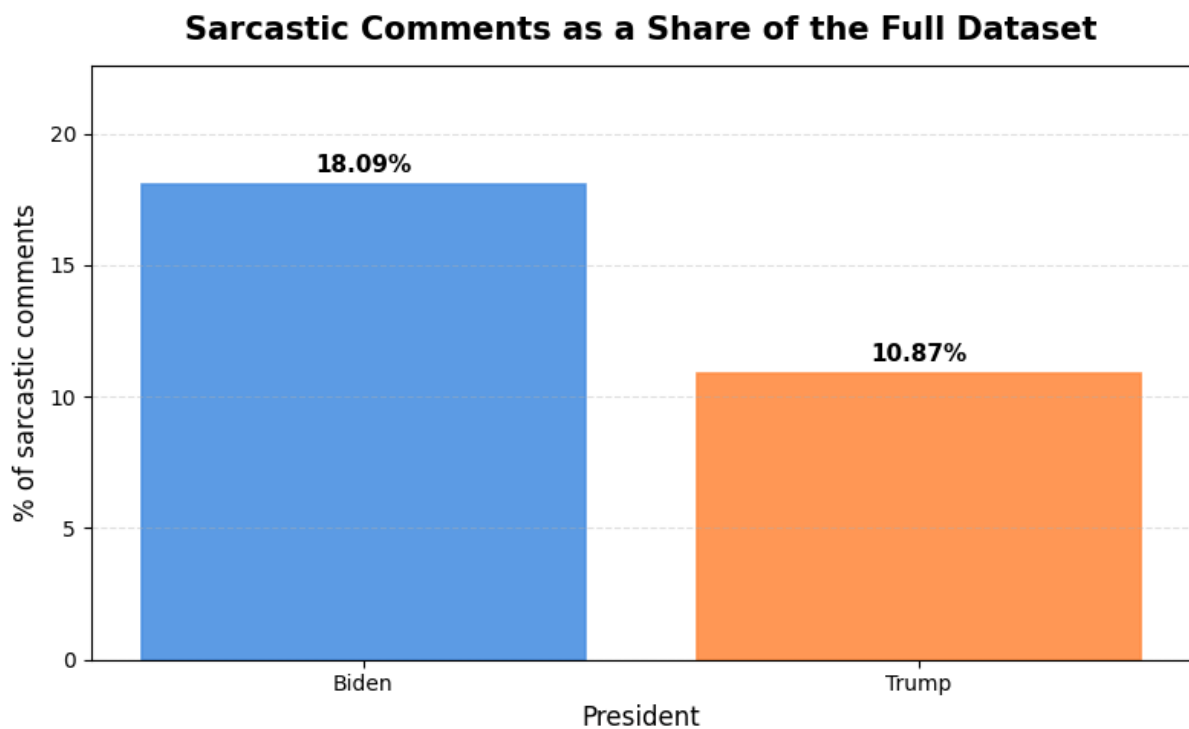


Figure 5. Trump-Biden share of sarcastic comments

Following **third objective**, K-means split videos in five distinct clusters in the following manner (*Fig.6*).

With regards to a *post-hoc semantic interpretation of clusters*, titles were based on the video names. It is worth mentioning the resulting classifications serve as heuristic interpretative tools and do not represent definitive thematic assignments. That is why we also formulated the global topics that align with suggested names of TF-IDF technique (Saqlain et al., 2016). A summary *Table 3* presents the cluster index, top three representative words suggesting cluster title, video examples and cluster sizes for analyzing the distribution of videos across the dataset.

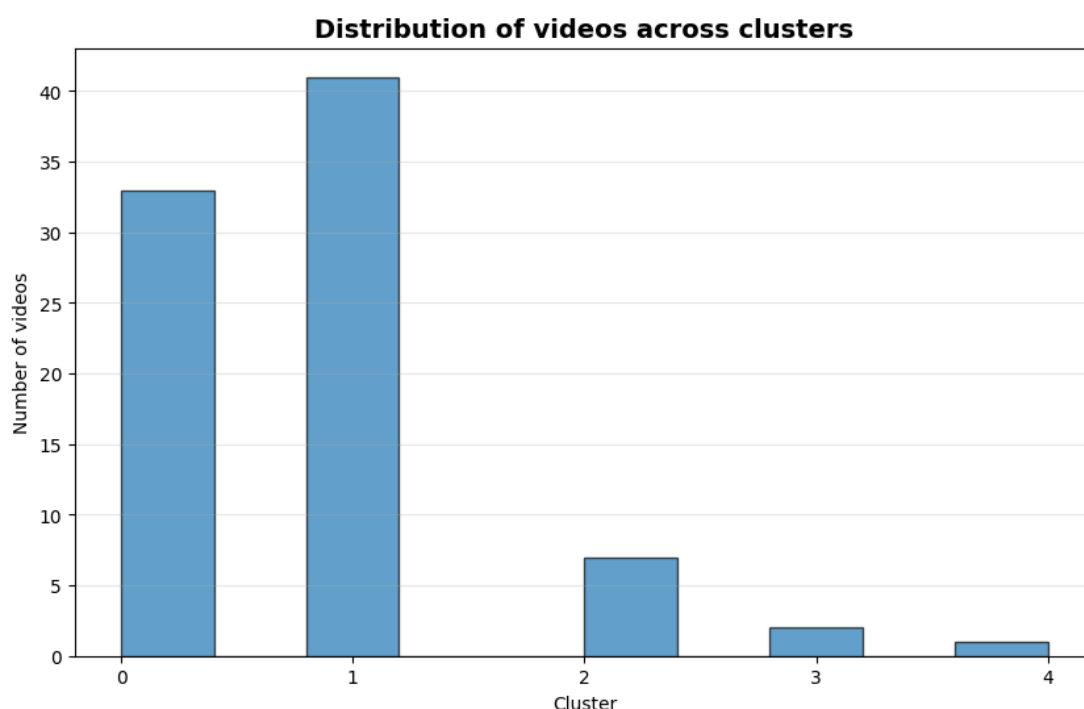


Figure 6. Number of videos across clusters

Table 3. Cluster labelling, both method- and human-inspected

Cluster	Top 3 words	Suggested Theme	Number of videos
0	shorts / biden / harris	Joe Biden agenda	33
1	agenda47 / america / trump	Electoral strategy & Agenda of Trump	41

2	harris 2024 / 2024 / biden harris	Biden-Harris 2024	7
3	republican shorts / republican / single republican	Republican campaign	2
4	israel shorts / stand israel / stand	Israel	1

Figure 7 represents the distribution of sarcastic comments across videos, where it is evident specific cluster of videos attract more than 30% of the audience exhibiting sarcasm. We question whether some specific videos are united in one group based on its sarcastic communication, hence illustrate it through K-means clustering.

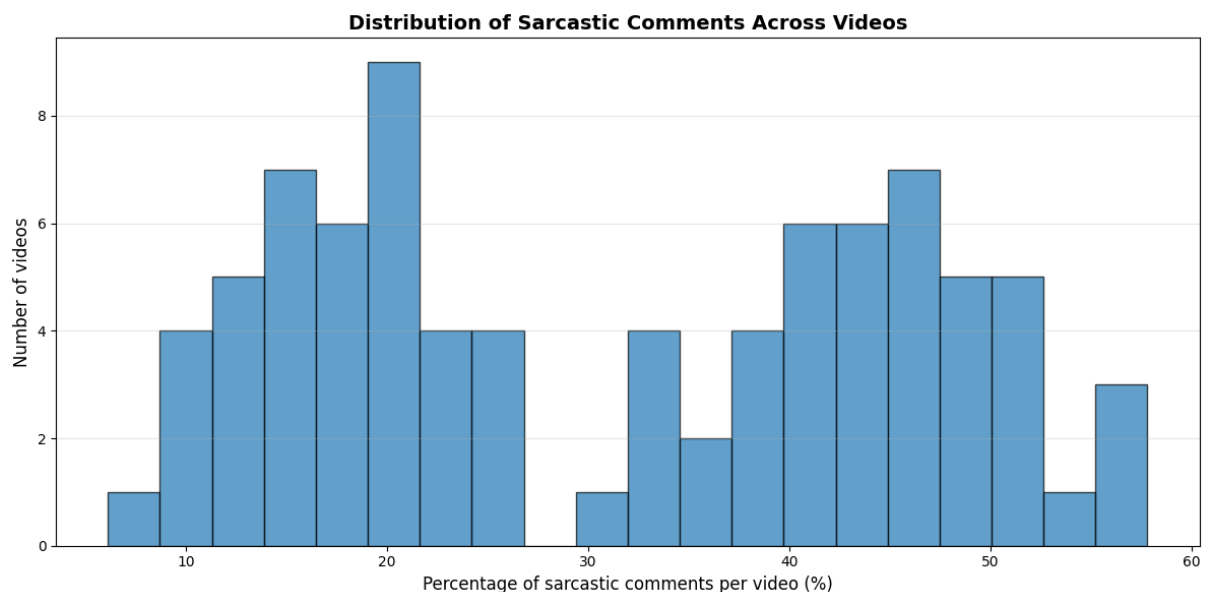


Figure 7. Distribution of sarcastic comments across videos

On figure 8 (a-b) the distribution of sarcastic comments across topic clusters is shown. The left panel displays the percentage of sarcastic comments per cluster, while the right panel shows the total number of comments associated with each cluster. Overall, these results demonstrate that sarcastic comments are unevenly distributed across topic clusters. Some clusters with relatively limited engagement display high sarcasm rates, while clusters with extensive engagement show lower proportions of sarcastic comments.

The highest proportion of sarcastic comments is observed in the Republican campaign cluster (48.1%), followed by the Joe Biden agenda cluster (45.4%) and the Biden–Harris 2024 cluster (39.4%). These clusters differ substantially in size: while the Joe Biden agenda cluster contains a large number of comments, the Republican campaign cluster includes a comparatively small number of observations.

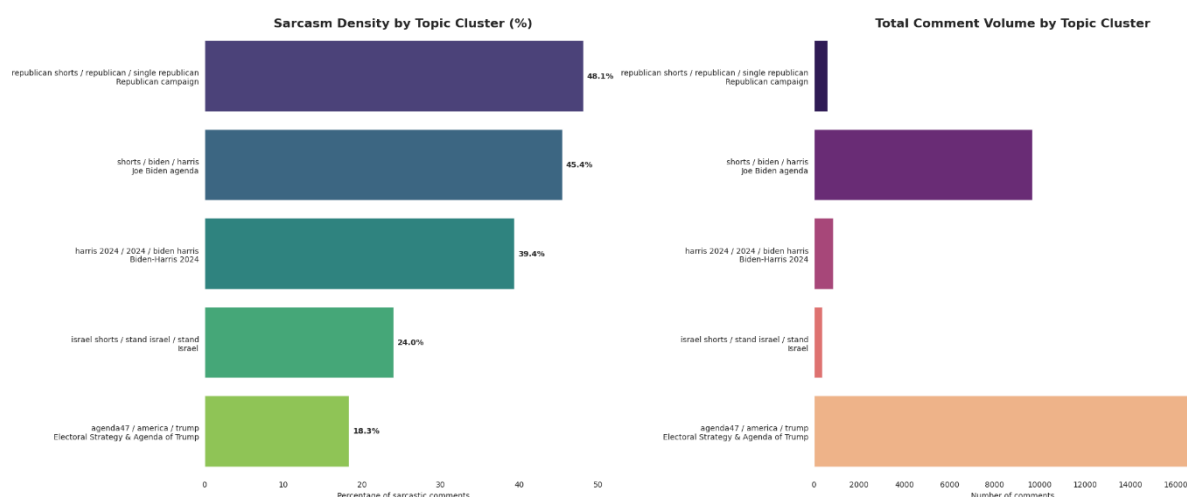


Figure 8(a-b). Sarcasm association with political topics, as well as its distribution

Lower sarcasm rates are found in clusters related to international affairs and campaign branding. Trump cluster exhibits the lowest sarcasm rate at 18.3%, despite accounting for the largest share of total comments ($N = 16,753$), which indicates that higher comment volume does not necessarily correspond to a higher proportion of sarcastic discourse. Answering **third objective**, electoral discussions of Biden and overall campaign-oriented discussions are the main sources of sarcasm, while geopolitically sensitive or politically related topics elicit comparatively fewer sarcastic reactions.

Discussion and Conclusion

This study examined sarcastic discourse in political YouTube comments related to Joe Biden and Donald Trump, with the aim of identifying which political topics are most strongly associated with sarcasm and whether these patterns differ between presidential candidates. By combining topic analysis with computational sarcasm detection, this research contributes to existing work on affective polarization and online political communication.

The findings indicate that sarcasm constitutes a substantial component of political discourse on YouTube. Approximately 29% of all analyzed comments were classified as sarcastic,

supporting earlier research that identifies sarcasm and mockery as common expressive strategies in highly polarized online environments (McCambridge, 2022; Nguyen et al., 2021). In line with theories of affective polarization, sarcastic language appears to function less as humor and more as a means of expressing political identity and emotional opposition (Bitecofer, 2020; Woods et al., 2024). However, this is also limited to zero-shot classifier technique which could have biased its split.

Importantly, sarcasm was not evenly distributed across political content. Certain videos attracted disproportionately high levels of sarcastic engagement, suggesting that sarcastic discourse clusters around specific topics. These topics are likely to be emotionally charged or identity-sensitive, such as leadership, legitimacy, or national values. This pattern aligns with previous research showing that political YouTube content is ideologically structured and shapes the tone of audience responses (Lai et al., 2024).

In line with research expectations, Biden-related videos exhibited a higher share of sarcastic comments. This asymmetry may reflect the continued memefication of Biden’s political persona, in which irony and sarcasm are frequently used to respond to his statements. Likewise, the observed pattern may be driven by growing dissatisfaction with Biden’s policies throughout his presidency (Tassadit, 2025). However, given the data manipulation strategies discussed in the wake of the Cambridge Analytica scandal (Arora et al., 2019), it is also possible that the Trump campaign encouraged the spread of positive comments, thereby toning down sarcastic content. Although such mechanisms go beyond our research scope, our results underline the importance of viewing online policy data as a dynamic and potentially manipulable environment when interpreting patterns of sarcasm and audience interaction.

From a methodological perspective, the comparison between RoBERTa-based supervised classification and zero-shot classification yielded important insights. Although RoBERTa achieved higher overall accuracy, its performance was limited by the relatively small size of the manually labeled training set, which reduced its ability to generalize to unseen data. In contrast, the zero-shot model demonstrated higher precision, recall, and F1 scores, indicating a stronger capacity to detect sarcastic language in a low-resource setting. This finding underscores the limitations of accuracy as a sole evaluation metric and supports the use of zero-shot approaches for large-scale sarcasm detection when labeled data are scarce (Yin et al., 2019).

However, our study had several limitations. First, pre-processing of the text could not capture all the “inside jokes” from YouTube comments and local themes such as “Bidenomics” that

originally pose a sarcastic meaning (Fowler, 2023). Additionally, even after several cleaning pipelines not all the typos or symbol decoding were correct, and not all the abbreviations were revealed, which could have weakened the labelling results. Secondly, approximately 20% of the data consisted of non-English comments. Systematic translation would have increased the number of observations and potentially identified additional instances of sarcasm. Thirdly, manual 500 comments categorization introduced an element of subjectivity, as sarcasm itself is a spectrum and can overlap with related rhetorical forms such as irony, satire, understatement, exaggeration, and rhetorical questions (Du et al., 2022). In general, manual labels may be interpreted as sarcastic by some audiences but not by others. In the same fashion, the use of pre-trained language models carries a risk of bias due to the data on which these models were initially trained, such as twitter posts of Zero-shot or context-specific references implied on roBERTa. Finally, the observed patterns in sarcasm may reflect broader dynamics, such as the spread of memes following a presidential speech, strategic manipulation of online discourse as discussed in the wake of the Cambridge Analytica scandal, or growing public dissatisfaction with political outcomes, rather than just the use of sarcasm (Arora et al, 2019).

In conclusion, this study demonstrates that sarcasm is a prominent and topic-dependent feature of political discourse on YouTube and reflects broader dynamics of affective polarization. Methodologically, it highlights the value of zero-shot classification for analyzing abstract pragmatic phenomena in large-scale social media data. Together, these findings contribute to a deeper understanding of how political polarization is expressed through language in contemporary online environments.

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Appendix

Table 4. Epochs trials through roBERTa model

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	No log	0.555	0.740000	0.000000	0.000000	0.000000
2	No log	0.543	0.740000	0.000000	0.000000	0.000000
3	No log	0.523	0.740000	0.000000	0.000000	0.000000